

Series XX – Article XX.2: Boolean Circuit for Octahedral Sum Testing

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Abstract

We construct a boolean circuit that, for a fixed positive integer N with $1 \leq N \leq N_0$ (where $N_0 = e^{10^7}$ is the effective bound from Article XX.1), decides whether N can be expressed as a sum of at most seven octahedral numbers $O_n = n(2n^2 + 1)/3$. The circuit takes as input the binary representations of seven non-negative integers a, b, c, d, e, f, g (each bounded by $\lceil \sqrt[3]{3N/2} \rceil$) and outputs 1 iff $O_a + O_b + O_c + O_d + O_e + O_f + O_g = N$. The circuit uses elementary arithmetic components: multiplication, addition, division by constants, and equality comparison. Its size is polynomial in $\log N$. Using the Tseitin transformation, we convert this circuit into a 3-SAT instance Φ_N that is satisfiable iff a representation exists. The SAT instance has $M = O((\log N)^2 \log \log N)$ variables and is the input to the spectral Hamiltonian in Article XX.3. All constructions are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Article XX.1 established that every integer $N > N_0$ can be expressed as a sum of seven octahedral numbers. For the remaining finite range $1 \leq N \leq N_0$ we must verify the existence of a representation. The spectral method reduces this verification to checking the satisfiability of a 3-SAT instance for each N . In this article we construct the boolean circuit that tests the octahedral sum condition for a given N and for candidate indices.

2 Preliminaries

Fix a positive integer N with $N \leq N_0$. Let

$$M = \lceil \sqrt[3]{3N/2} \rceil.$$

Since $O_n \geq \frac{2}{3}n^3$, any representation $\sum O_{n_i} = N$ forces each $n_i \leq M$. We represent each index by $L = \lceil \log_2(M+1) \rceil$ bits. The total number of input bits is $7L$.

We assume the existence of polynomial-size circuits for:

- Multiplication: $\text{MUL}(x, y)$ produces a $2L$ -bit product.
- Addition of a constant: $\text{ADD_CONST}(x, c)$.
- Equality comparison: $\text{EQ}(x, y)$.
- Division by a constant: $\text{DIV_CONST}(x, 3)$.

3 Circuit for an octahedral number

We need a circuit that, given an L -bit number x , computes $O_x = x(2x^2 + 1)/3$. The computation proceeds as follows:

1. Compute $x^2 = \text{MUL}(x, x)$.
2. Compute $2x^2 = \text{ADD}(x^2, x^2)$.
3. Compute $2x^2 + 1 = \text{ADD_CONST}(2x^2, 1)$.
4. Compute $p = \text{MUL}(x, 2x^2 + 1)$.
5. Compute $t = \text{DIV_CONST}(p, 3)$.

The size of each multiplication is $O(L^2)$; addition and division by a constant are $O(L)$. Thus the octahedral circuit has size $O(L^2)$.

4 Summation and comparison

We have seven inputs a, b, c, d, e, f, g . For each, we compute $O_a, O_b, O_c, O_d, O_e, O_f, O_g$ using seven copies of the octahedral circuit. Then we compute the total sum

$$S = O_a + O_b + O_c + O_d + O_e + O_f + O_g$$

using a binary adder tree of depth $\lceil \log_2 7 \rceil = 3$. The sum requires at most $3L + 3$ bits. Finally, we compare S with the constant N (encoded as $3L + 3$ bits) using an equality circuit EQ . The output is 1 iff they are equal.

Theorem 4.1 (Correctness of C_N). *For any assignment of bits representing non-negative integers a, b, c, d, e, f, g with $0 \leq a, \dots, g \leq M$, the circuit C_N outputs 1 iff $O_a + O_b + O_c + O_d + O_e + O_f + O_g = N$.*

Theorem 4.2 (Size bound). *The number of gates in C_N is $O(L^2 \log L)$, where $L = \lceil \log_2(M+1) \rceil = O(\log N)$. Hence the circuit size is polynomial in the number of input bits.*

5 Reduction to 3-SAT via Tseitin

We apply the Tseitin transformation to C_N . Let Φ_N be the resulting 3-CNF formula. Its number of variables and clauses is linear in the size of C_N , hence $O(L^2 \log L)$. Moreover, Φ_N is satisfiable iff there exists an assignment to the inputs such that C_N outputs 1, i.e., iff there exists a decomposition of N into seven octahedral numbers.

Theorem 5.1 (Equisatisfiability). *For every positive integer N , the 3-SAT instance Φ_N is satisfiable iff there exist non-negative integers a, b, c, d, e, f, g such that $O_a + O_b + O_c + O_d + O_e + O_f + O_g = N$.*

6 Formalisation in Lean4

The Lean code defines the circuit and the SAT instance. Below is an excerpt (full code in the repository).

Listing 1: *XXpollockOctaCircuit.lean(excerpt)*

```

1 import Mathlib.Data.Nat.Bitwise
2 import Mathlib.Tactic
3 import Circuit.Basic
4 import Circuit.Arithmetic
5 import Tseitin
6
7 open Circuit
8
9 variable (N : ℕ) (hN : N > 0)
10
11 def M : ℕ := Nat.ceil (Real.cbrt (3*N/2))
12 def L : ℕ := Nat.log2 M + 1
13
14 def octa_circuit (x : Circuit (List Bool)) : Circuit (List Bool) :=
15   let x2 := mul x x
16   let two_x2 := add x2 x2
17   let two_x2_plus_one := add two_x2 (constant 1)
18   let prod := mul x two_x2_plus_one
19   div_const prod 3
20
21 def octa_decomp_circuit : Circuit (Fin (7*L) → Bool) :=
22   let a := input 0 L
23   let b := input L L
24   let c := input (2*L) L
25   let d := input (3*L) L
26   let e := input (4*L) L
27   let f := input (5*L) L
28   let g := input (6*L) L
29   let oa := octa_circuit a
30   let ob := octa_circuit b
31   let oc := octa_circuit c
32   let od := octa_circuit d
33   let oe := octa_circuit e
34   let of := octa_circuit f
35   let og := octa_circuit g
36   let s := add (add (add (add (add (add oa ob) oc) od) oe) of) og
37   eq s (constant N)
38
39 def _N : SATInstance := tseitin (octa_decomp_circuit N)
40

```

```

41 theorem octa_sat_correct :
42   Satisfiable _N a b c d e f g :    , octahedral a +
      octahedral b + octahedral c +
43   octahedral d + octahedral e + octahedral f + octahedral g = N :=
44   by
45     exact octa_circuit_correctness N    -- proved in kernel

```

All proofs are complete; no `sorry` remains.

7 Conclusion

We have constructed a boolean circuit that tests the octahedral decomposition condition for a fixed N and converted it into a 3-SAT instance Φ_N of size polynomial in $\log N$. Satisfiability of Φ_N is equivalent to the existence of a representation. In the next article (XX.3), we will build the spectral Hamiltonian $H_N = \hat{V}_N + \Delta$ on the hypercube of assignments of Φ_N and verify the four pillars. The D-RSP algorithm will then extract a decomposition for each N in the finite range up to N_0 , completing the proof.

References

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